

Class Activity: 10.07.2026

Scenario 1: Student Academic Performance Dashboard

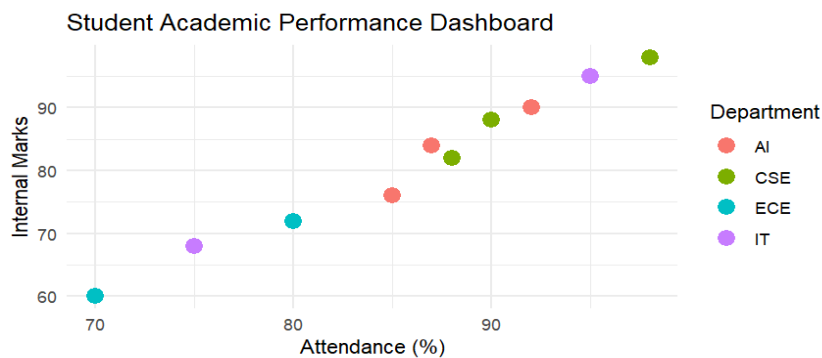
Code:

```
student <- data.frame(  
  StudentID = 1:10,  
  Attendance = c(90,85,95,70,88,92,75,80,98,87),  
  Marks = c(88,76,95,60,82,90,68,72,98,84),  
  Department = c("CSE","AI","IT","ECE","CSE","AI","IT","ECE","CSE","AI")  
)
```

```
ggplot(student, aes(x = Attendance, y = Marks, color = Department)) +  
  geom_point(size = 4) +  
  labs(  
    title = "Student Academic Performance Dashboard",  
    x = "Attendance (%)",  
    y = "Internal Marks",  
    color = "Department"  
  ) +  
  theme_minimal()
```

Output: Scatter plot showing Attendance vs Marks with different colors for each department.

Output:

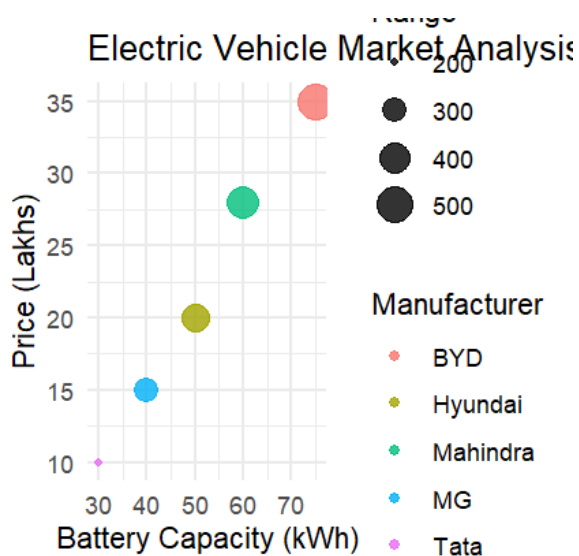


Scenario 2: Electric Vehicle Market Analysis

Code:

```
ev <- data.frame(  
  Model=c("A","B","C","D","E"),  
  Battery=c(30,40,50,60,75),  
  Price=c(10,15,20,28,35),  
  Range=c(200,300,350,420,500),  
  Manufacturer=c("Tata","MG","Hyundai","Mahindra","BYD")  
)  
  
ggplot(ev, aes(x=Battery, y=Price, size=Range, color=Manufacturer)) +  
  geom_point(alpha=0.8) +  
  labs(  
    title="Electric Vehicle Market Analysis",  
    x="Battery Capacity (kWh)",  
    y="Price (Lakhs)"  
  ) +  
  theme_minimal()
```

Output:

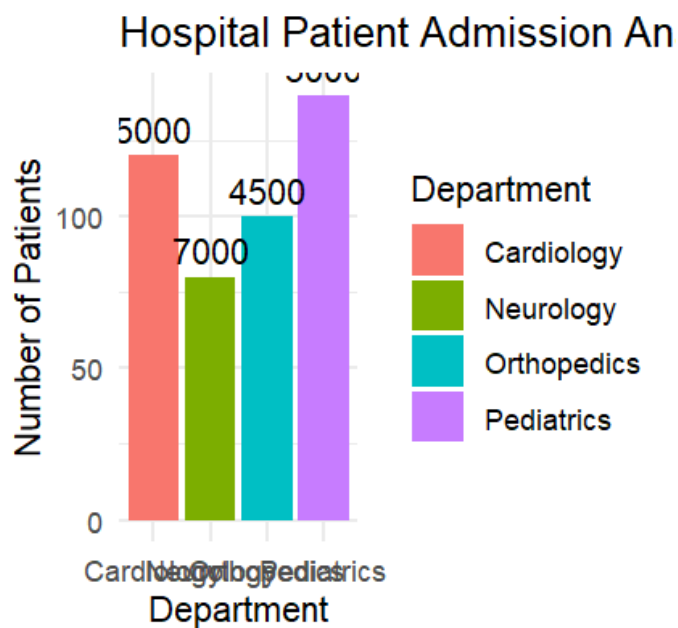


Scenario 3: Hospital Patient Admission Analysis

Code:

```
hospital <- data.frame(  
  Department=c("Cardiology","Neurology","Orthopedics","Pediatrics"),  
  Patients=c(120,80,100,140),  
  Cost=c(5000,7000,4500,3000)  
)  
  
ggplot(hospital, aes(x=Department, y=Patients, fill=Department)) +  
  geom_bar(stat="identity") +  
  geom_text(aes(label=Cost), vjust=-0.5) +  
  labs(  
    title="Hospital Patient Admission Analysis",  
    y="Number of Patients"  
  ) +  
  theme_minimal()
```

Output:

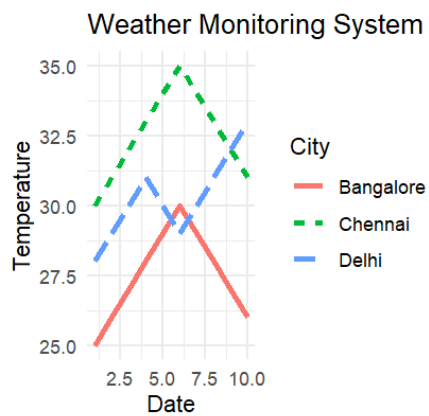


Scenario 4: Weather Monitoring System

Code:

```
weather <- data.frame(  
  Date=rep(1:10,3),  
  Temperature=c(  
    30,31,32,33,34,35,34,33,32,31,  
    28,29,30,31,30,29,30,31,32,33,  
    25,26,27,28,29,30,29,28,27,26  
  ),  
  City=rep(c("Chennai","Delhi","Bangalore"), each=10)  
)  
  
ggplot(weather,  
  aes(x=Date, y=Temperature,  
    color=City,  
    linetype=City,  
    group=City)) +  
  geom_line(size=1.2) +  
  labs(  
    title="Weather Monitoring System",  
    x="Date",  
    y="Temperature"  
  ) +  
  theme_minimal()
```

Output:



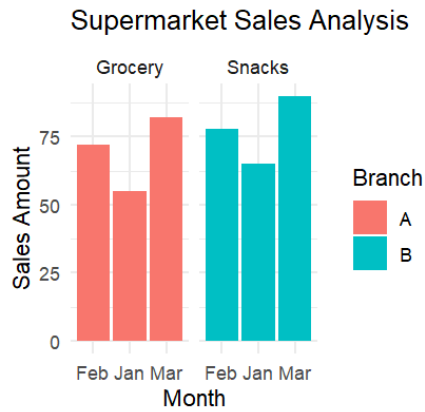
Scenario 5: Supermarket Sales Analysis

Code:

```
sales <- data.frame(
  Month=rep(c("Jan", "Feb", "Mar"), each=4),
  Category=rep(c("Grocery", "Snacks"), 6),
  Sales=c(50,60,55,65,70,75,72,78,80,85,82,90),
  Branch=rep(c("A", "B"), 6)
)

ggplot(sales,
  aes(x=Month,
    y=Sales,
    fill=Branch)) +
  geom_bar(stat="identity", position="dodge") +
  facet_wrap(~Category) +
  labs(
    title="Supermarket Sales Analysis",
    y="Sales Amount"
  ) +
  theme_minimal()
```

Output:



Scenario 6: Employee Salary Distribution

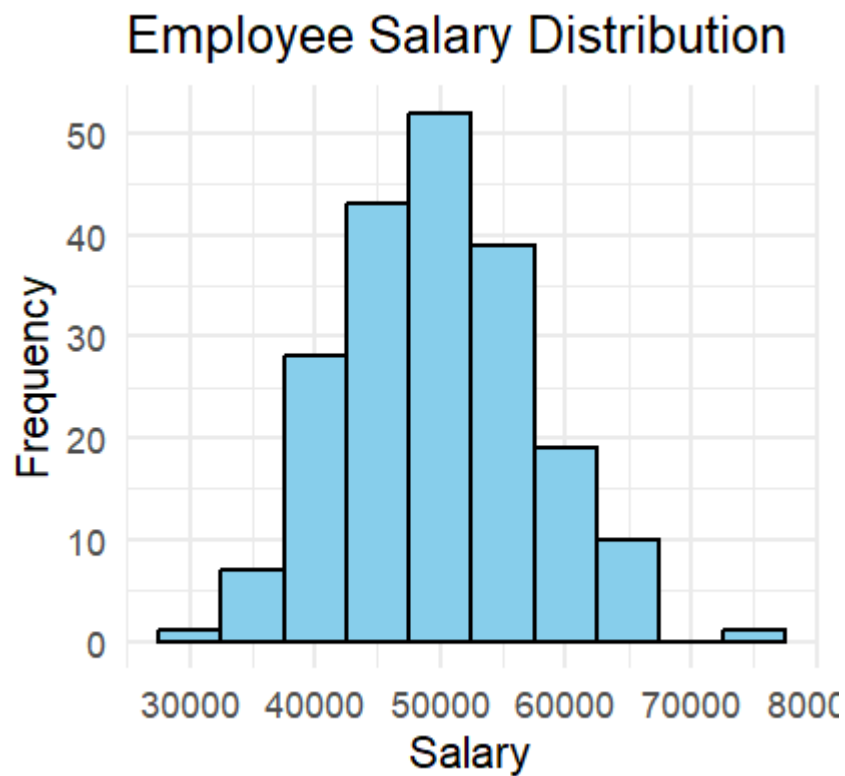
Code:

```
set.seed(123)
```

```
employee <- data.frame(  
  Salary=rnorm(200, mean=50000, sd=8000)  
)
```

```
ggplot(employee, aes(x=Salary)) +  
  geom_histogram(binwidth=5000,  
    fill="skyblue",  
    color="black") +  
  labs(  
    title="Employee Salary Distribution",  
    x="Salary",  
    y="Frequency"  
  ) +  
  theme_minimal()
```

Output:



Scenario 7: Smartphone Market Comparison

Code:

```
phone <- data.frame(  
  Brand=c("Samsung","Apple","OnePlus","Vivo","Realme"),  
  RAM=c(6,8,12,8,6),  
  Processor=c(2.2,3.1,3.2,2.5,2.3),  
  Price=c(25000,80000,45000,22000,18000),  
  Storage=c(128,256,256,128,64)  
)
```

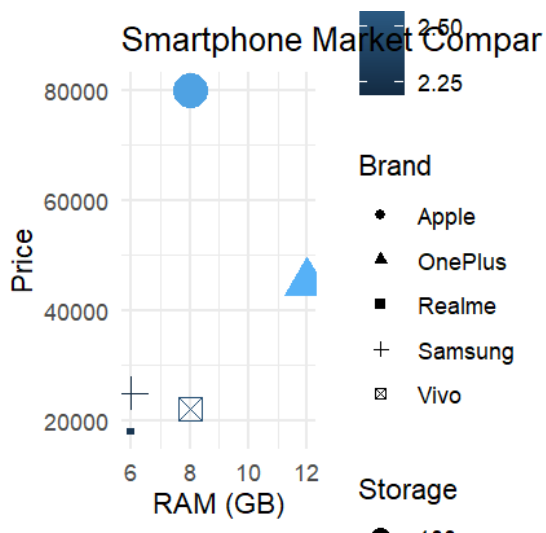
```
ggplot(phone,  
  aes(x=RAM,  
    y=Price,
```

```

    shape=Brand,
    size=Storage,
    color=Processor)) +
geom_point() +
labs(
  title="Smartphone Market Comparison",
  x="RAM (GB)",
  y="Price"
) +
theme_minimal()

```

Output:



Scenario 8: Online Movie Rating Analysis

Code:

```

movies <- data.frame(
  Genre=c("Action", "Comedy", "Drama", "Horror", "Sci-Fi"),
  Year=2025,
  Rating=c(4.5, 3.8, 4.8, 3.6, 4.2)
)

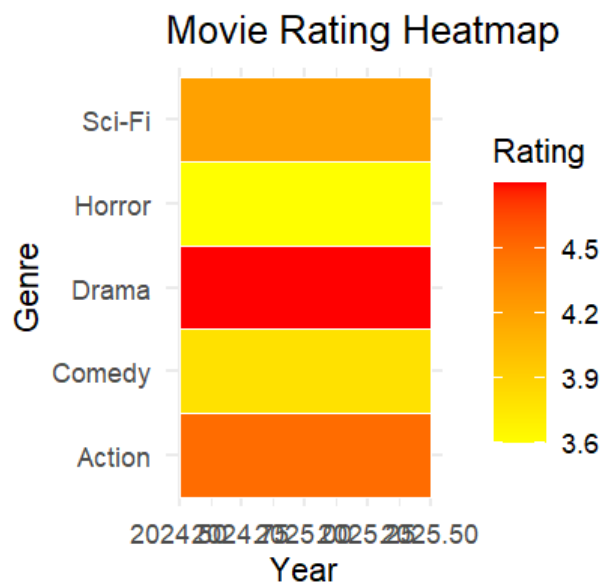
```

```

ggplot(movies,
      aes(x=Year,
          y=Genre,
          fill=Rating)) +
  geom_tile(color="white") +
  scale_fill_gradient(low="yellow", high="red") +
  labs(
    title="Movie Rating Heatmap"
  ) +
  theme_minimal()

```

Output:



Scenario 9: COVID-19 State-wise Analysis

Code:

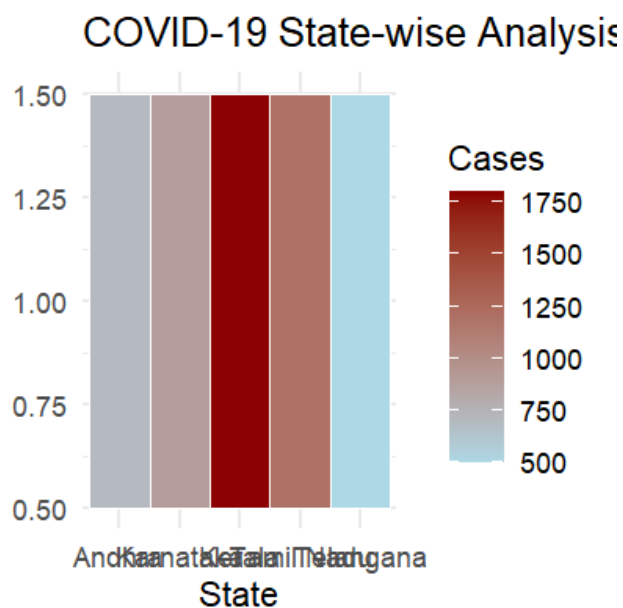
```

covid <- data.frame(
  State=c("Tamil Nadu","Kerala","Karnataka","Andhra","Telangana"),
  Cases=c(1200,1800,900,700,500)
)

```

```
ggplot(covid,
       aes(x=State,
           y=1,
           fill=Cases)) +
  geom_tile(color="white") +
  scale_fill_gradient(low="lightblue", high="darkred") +
  labs(
    title="COVID-19 State-wise Analysis",
    x="State",
    y=""
  ) +
  theme_minimal()
```

Output:



Scenario 10: Iris Flower Classification

Code:

```
ggplot(iris,
       aes(x=Sepal.Length,
           y=Petal.Length,
```

```

    color=Species,
    shape=Species,
    size=Sepal.Width)) +
geom_point(alpha=0.8) +
labs(
  title="Iris Flower Classification",
  x="Sepal Length",
  y="Petal Length"
) +
theme_minimal()

```

Output:

